

MODULE 08 SUMMARY

Polynomial and Rational Expressions

Big Picture

Module 08 focused on structural algebra. You learned how to factor polynomial expressions, simplify rational expressions, perform operations with rational expressions, simplify complex fractions, divide polynomials using multiple methods, and evaluate limits graphically and algebraically.

Every skill in this module builds on structure recognition. Factoring exposes hidden components. Rational expressions rely on factoring. Complex fractions require structural rewriting. Polynomial division reveals quotient and remainder structure. Limits rely on understanding removable and infinite discontinuities.

Essential Standard

Students will factor polynomial expressions, perform operations with rational expressions, simplify complex fractions, and divide polynomials, connecting algebraic structure to procedural fluency while applying mathematical practices to model real-world situations and extend learning through limits.

Top 10 Core Concepts

Concept	Description
1. Greatest Common Factor	Always factor out the GCF before applying other methods.
2. Difference of Squares	$a^2 - b^2 = (a - b)(a + b)$
3. Sum and Difference of Cubes	$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$
4. Canceling Factors	Only common factors may cancel in rational expressions.
5. Restrictions	Denominators cannot equal zero; state restrictions clearly.
6. Reciprocal Rule	Division of rational expressions becomes multiplication by the reciprocal.
7. Complex Fraction Strategy	Rewrite using reciprocal or multiply by the LCD.
8. Long Division	Divide leading terms, multiply, subtract, repeat.
9. Synthetic Division	Shortcut for divisors of the form $x - c$.
10. Removable vs Infinite Discontinuity	Canceling factors create holes; uncanceled denominators create asymptotes.

Section 8.A — Factor Expressions

Key Structures

$$a^2 - b^2 = (a - b)(a + b)$$
$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$
$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Sample Questions

- Factor: $12x^2 - 18x$
- Factor: $x^2 + 7x + 10$
- Factor completely: $6x^3 - 54x$
- Factor completely: $4x^3 + 16x^2 - 9x - 36$
- Explain why factoring must begin with identifying the GCF.

Section 8.B — Rational Expressions

Key Ideas

- Factor before canceling.
- State restrictions.
- Use common denominators.

Sample Questions

- Simplify: $\frac{x^2-16}{x-4}$
- Multiply: $\frac{x}{x+2} \cdot \frac{x+2}{3}$
- Add: $\frac{1}{x} + \frac{2}{x-1}$
- Simplify: $\frac{x^2-4}{x^2-2x}$
- Explain why canceled factors still create restrictions.

Section 8.C — Complex Fractions

Key Strategies

- Rewrite using reciprocal.
- Multiply by LCD.
- Factor completely.

Sample Questions

- Simplify: $\frac{\frac{x+1}{x}}{\frac{1}{x}}$
- Simplify: $\frac{\frac{1}{x} + \frac{1}{x+1}}{\frac{1}{x}}$
- Simplify: $\frac{\frac{x^2-4}{x-2}}{\frac{x}{x-2}}$
- Simplify: $\frac{\frac{3}{x} - \frac{1}{x}}{\frac{x}{x}}$
- Explain why the reciprocal method works.

Section 8.D — Divide Polynomials

Key Methods

- Long division
- Box method
- Synthetic division

Sample Questions

- Divide: $\frac{x^3-2x^2+4x-8}{x-2}$
- Divide using box method: $\frac{2x^3+3x^2-5x+6}{x+2}$
- Divide using synthetic division: $\frac{x^4-3x^3+2x-5}{x-1}$
- Divide: $\frac{x^2+5x+6}{x+2}$
- Explain why zero placeholders are necessary.

Section 8.E — Limits and Discontinuities

Key Concepts

- One-sided limits
- Removable discontinuity
- Infinite discontinuity
- Jump discontinuity

Sample Questions

- Evaluate: $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2}$
- Evaluate: $\lim_{x \rightarrow 3} \frac{1}{x-3}$
- Classify: $\frac{x^2-4}{x-2}$
- Evaluate: $\lim_{x \rightarrow -2} \frac{x^2-4}{x+2}$
- Explain why jump discontinuities prevent two-sided limits.